CHRIST (Deemed to be University)

School of Business and Management

Pune, Lavasa Campus

Master of Business Administration (MBA)

Batch 2025-27

Trimester IV

Real-Time Short-Term Rental Analytics for Airbnb

An ELT Pipeline, SQL KPI Engine and Power BI Dashboard Comparing New York and London

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Class	4MBA BA PLC
Course Code & Name	BUSINESS DECISION INTELLIGENCE - MBA443B
Assessment	Individual Analytics Project: Dashboard Development and SQL Practical
Campus of Study	Pune Lavasa Campus
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Date	11-10-2026

Abstract

This report documents an individual analytics project built around a real business scenario: how Airbnb, a company whose data platform was built on Hadoop, can monitor and compare two of its largest city markets, New York and London. Public Inside Airbnb snapshots from September 2026 (121,872 listings and 3.32 million reviews) were loaded into MySQL through an ELT pipeline. Raw files land in staging tables first, and twelve logged cleaning rules then turn them into a star schema with one fact table for listings and one for reviews. Sixteen analytical queries use joins, common table expressions, window functions and prepared statements, and three stored procedures provide dynamic filters. A scheduled MySQL event captures a KPI snapshot every 15 minutes, and Power BI reads the semantic views through the MySQL connector. The dashboard has four pages with slicers, a drill-down matrix, a drill-through page and gauges that track KPIs against targets. The analysis shows two very different markets. London has three times the supply and 43.3% of its listings are active, against 31.0% in New York. After Local Law 18 took effect in September 2023, 80.7% of New York listings now require stays of 30 nights or more, and twelve-month review volume fell by a third before it began to recover. In both cities the largest 10% of hosts control about 45% of listings. The report closes with a short comparison against Booking Holdings and Expedia and with eight prioritised recommendations for city managers.

Keywords: ELT, MySQL, Power BI, KPI tracking, drill-through, short-term rentals, Airbnb, Hadoop

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List of Abbreviations

Table 1: Abbreviations and terms

Term	Meaning
BI	Business intelligence
CTE	Common table expression (WITH clause in SQL)
DAX	Data Analysis Expressions, the Power BI formula language
ELT / ETL	Extract, load, transform / extract, transform, load
FX	Foreign exchange rate (GBP to USD = 1.3371)
GBV	Gross booking value
HDFS	Hadoop Distributed File System
KPI	Key performance indicator
LL18	New York City Local Law 18 of 2022 (short-term rental registration)
LTM	Last twelve months before the data snapshot
NYC / LON	New York City / London (city codes used in the database)
OSE	NYC Mayor's Office of Special Enforcement
P99	99th percentile, used as the price outlier cut-off
STR	Short-term rental (stay of fewer than 30 nights)

1. Introduction

1.1 Background and Motivation

Airbnb does not own a single room, yet on any night it is one of the largest providers of accommodation in the world. Its business depends on two groups of people, hosts who list homes and guests who book them, and on a steady flow of data about prices, availability, reviews and local rules. When a city changes its regulations, the effect shows up first in that data: listings switch to longer minimum stays, reviews slow down and some hosts leave. A city manager who can see these signals early can act before revenue is lost. Research also links home-sharing to higher rents and lost hotel revenue, which is why cities keep tightening the rules (Barron et al., 2021; Wachsmuth & Weisler, 2018; Zervas et al., 2017).

I chose Airbnb for three reasons. First, the company is a well documented Hadoop user. Its engineering team has described a platform built on HDFS, Hive, Presto and Spark, and it created Apache Airflow to schedule those pipelines (Mayfield et al., 2016; Giamas, 2015). Second, detailed listing and review data for individual cities is published by Inside Airbnb, so the work can be repeated by anyone. Third, New York and London offer a natural experiment. New York began enforcing Local Law 18 on 5 September 2023 (NYC Mayor's Office of Special Enforcement, n.d.), while London has kept its 90-night annual cap on whole-home lets (UK Parliament, 2015).

1.2 Business Scenario and Problem Statement

The scenario is written from the point of view of a regional strategy team at Airbnb that looks after North America and the United Kingdom. The team needs one place to answer a recurring question: are our two biggest English-speaking city markets healthy, and where should we put effort next? In concrete terms the team wants to know how much supply each city has, how much of it is genuinely active, how regulation is changing guest demand, which hosts and neighbourhoods carry the business, and how the picture changes as new data arrives. The problem is therefore both technical, because the data has to be cleaned, modelled and refreshed, and managerial, because the output has to support decisions.

1.3 Objectives and Research Questions

Table 2: Research questions and where they are answered

No.	Research question	Method	Section
RQ1	How can raw city data be turned into a reliable, refreshable analytics store?	ELT pipeline, star schema, cleaning log	3
RQ2	Which KPIs describe the health of a short-term rental market, and how do the two cities compare?	Advanced SQL, KPI framework	4, 6

No.	Research question	Method	Section
RQ3	How can managers explore the KPIs themselves and see them update?	Power BI with filters, drill-down, drill-through, refresh	5
RQ4	What has regulation done to supply and demand in New York?	Review trend index, stay-policy split	6.2
RQ5	How does Airbnb compare with its main online competitors?	Annual report comparison	7
RQ6	What should city managers do next?	Prioritised recommendations	8

2. Data and Big Data Context

2.1 Data Sources

All data comes from Inside Airbnb, a not-for-profit project that publishes scraped Airbnb data for research and public debate. I used the September 2026 snapshots, taken on 14 September for New York and 17 September for London. London prices are in pounds and were converted to US dollars at 1.3371, the mid-rate on the London snapshot date. The calendar files hold 44.5 million daily rows; they were downloaded and checked, but the final build leaves them out because the review data already carries the demand signal and the calendar made a laptop build take well over ten minutes instead of about four.

Table 3: Datasets used

File	New York	London	Key fields
listings.csv	29,918 rows	91,954 rows	listing, host, neighbourhood, room type, price, minimum nights, reviews, licence
reviews.csv	999,099 rows	2,323,969 rows	listing id, review date
neighbourhoods.csv	230 rows	33 rows	neighbourhood and borough lookup
calendar.csv (not loaded)	10.9 million rows	33.6 million rows	date, available flag, price

Source: Inside Airbnb (2026); row counts from the MySQL load log.

The complete pipeline was run and checked on the full dataset. For the live demonstration on a student laptop, where MySQL kept stalling on the 2.3 million London reviews, the same scripts were pointed at a random sample of 2,000 listings per city. The sample reproduces the full-data ratios closely (for example, an active share of 43.5% against 43.3% in London and a long-stay share of 81.1% against 80.7% in New York), so the dashboard behaves like the full model. All findings in Chapter 6 use the full dataset.

2.2 Hadoop and Cloud Context

At Airbnb the equivalent of this project runs at a much larger scale. Event data reaches the warehouse through Kafka, production databases are copied in with Sqoop, Hive tables on HDFS act as the single source of truth, Presto serves ad hoc queries and Spark handles machine learning and streaming (Mayfield et al., 2016). Airflow, which Airbnb wrote and later gave to the Apache Foundation, schedules the jobs. My project keeps the same layers

but swaps each one for a tool that fits on a laptop, as Table 4 shows. Because every layer is plain SQL, moving it to a cloud warehouse would mostly mean changing where the files are stored and what triggers the jobs, not rewriting the logic.

Table 4: How the project maps onto a Hadoop and cloud architecture

Layer	This project	Airbnb-style big data / cloud equivalent
Storage of raw files	CSV files in the MySQL upload folder	HDFS or Amazon S3 data lake
Staging (raw tables)	stg_listings, stg_reviews	Hive external tables over raw partitions
Transformation	SQL in MySQL (scripts 02 and 03)	Hive or Spark SQL jobs
Scheduling	MySQL EVENT every 15 minutes	Apache Airflow DAG
Serving layer	vw_pbi_* views	Presto or a cloud warehouse such as Azure Synapse
Dashboard refresh	Power BI Desktop, Import mode, manual or scheduled refresh	Power BI Service with a gateway and scheduled or DirectQuery refresh

3. ETL/ELT Workflow

3.1 Why ELT and Not ETL

In a classic ETL design the data is cleaned on the way in, so the warehouse only ever sees the finished version. In ELT the raw files are loaded first and the cleaning happens inside the database. I chose ELT for two reasons. The raw rows stay available, so any cleaning rule can be checked or changed and re-run without downloading the files again. And the transformation is written in SQL, which is easier to read and audit than a chain of spreadsheet steps. Every business assumption, such as the exchange rate, the review rate and the occupancy cap, sits in a small parameter table rather than inside the code, so a manager can change an assumption and rebuild the model with one procedure call.

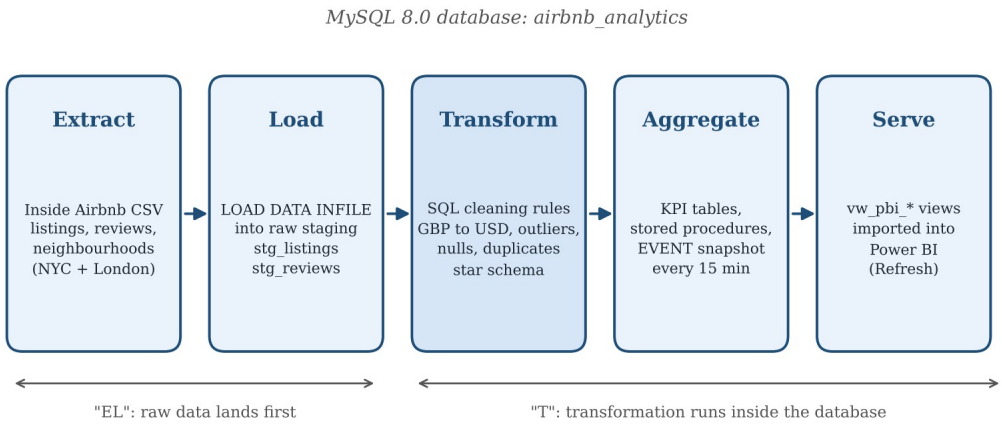


Figure 1: ELT workflow from raw files to the Power BI dashboard

3.2 Extract and Load

The six CSV files are loaded with the server-side LOAD DATA INFILE command (Oracle, 2025) into staging tables that mirror the raw columns. Nothing is cleaned at this stage; quoted text, line breaks inside listing names and missing values are kept as they are. Each load writes its row count to an audit table, etl_run_log, so the first thing anyone sees after a run is whether the expected 121,872 listings and 3.32 million reviews arrived. An early version used the LOCAL variant of the command, which MySQL Workbench blocks by default; switching to the server-side form and the secure upload folder solved errors 3948 and 2068 and made the load repeatable.

3.3 Transform: Cleaning Rules and Star Schema

Twelve rules turn the staging data into analysis-ready tables. A key design choice was to flag rather than delete. For example, about 35% of listings have no price in the snapshot. Deleting them would have hidden a third of the supply, so they are kept with a price_status of Missing and simply left out of price and revenue KPIs. Every rule writes how many rows it touched to data_quality_log, which feeds the data quality view in Power BI.

Table 5: Main cleaning rules and their effect on the full dataset

Rule	What it checks	Action	New York	London
R1	Missing price	Kept, flagged, excluded from price KPIs	10,512	32,049
R2	Price above city 99th percentile	Kept, flagged as outlier	194	599
R4	Missing minimum nights	Filled with the city median	8	309
R5	Minimum nights above 365	Capped at 365	16	24
R6	No reviews per month (never reviewed)	Set to 0	8,114	20,694
R9	Line breaks or extra spaces in names	Collapsed to single spaces	841	1,596
R12	Same listing reviewed twice on one day	Kept, as these are genuine	37,505	20,805

Source: data_quality_log table after the full run. Rules R3, R7, R8, R10 and R11 found no problem rows.

The cleaned data is organised as a star schema (Kimball & Ross, 2013). fact_listing holds one row per listing with derived measures such as estimated occupancy and revenue, and fact_review holds one row per review. Five dimensions, for city, neighbourhood, room type, host and date, surround them. Estimated occupancy follows the Inside Airbnb method: reviews in the last twelve months are divided by a 50% review rate, multiplied by an average stay of three nights or the minimum stay if that is longer, and capped at 70% (Inside Airbnb, n.d.).

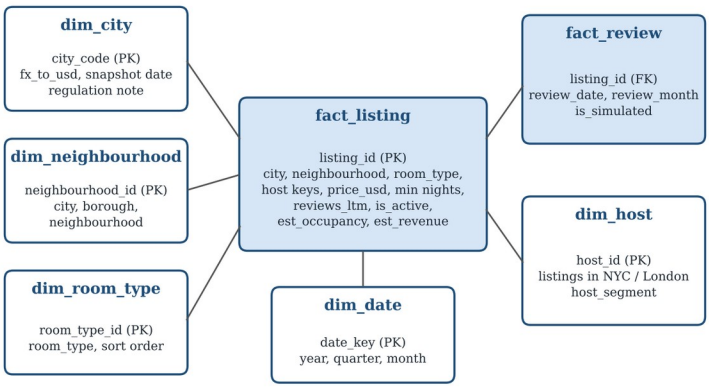


Figure 2: Star schema of the airbnb_analytics database

3.4 The Real-Time Refresh Engine

Real-time in this project means near-real-time micro-batching, which is how most business dashboards actually work. Three pieces make it run. A stored procedure, sp_refresh_all, rebuilds the monthly review aggregates. A second procedure, sp_capture_kpi_snapshot, writes the current KPIs for each city with a timestamp into kpi_snapshot. Two MySQL events call these on a timer: one every 15 minutes and a full refresh every night at 02:00. To show the flow during the viva, sp_simulate_live_reviews inserts new reviews marked as simulated, refreshes the aggregates and takes a snapshot. When the Power BI report is refreshed, the new snapshot row and the new review counts appear straight away. Simulated rows can be removed with one call, so the demonstration never contaminates the real data.

4. SQL Analytics and KPI Tracking

4.1 Advanced SQL Techniques

Script 05 holds sixteen analytical queries, each written to answer a single business question rather than to show off syntax. Table 6 lists the main techniques and where they were useful.

Table 6: SQL techniques and the business questions they answer

Technique	Example in the project	Business question
Multi-table joins	fact_listing joined to city, neighbourhood, room type and host dimensions	Which boroughs earn the most?
Conditional aggregation (pivot)	SUM(city_code = 'NYC') beside SUM(city_code = 'LON')	How does the room mix differ by city?
CTEs with window functions	ROW_NUMBER and COUNT over a partition to find the median price	What is a typical nightly price?
LAG and running totals	Year-on-year review growth and a twelve-month rolling sum	How did demand react to Local Law 18?
NTILE	Hosts ranked into percentiles; prices into quartiles	Do a few hosts dominate supply?
Pareto with cumulative share	Neighbourhoods sorted by revenue with a running percentage	How concentrated is revenue?
Prepared statements	Parameters for city, room type and price range	Same report, any slice

Technique	Example in the project	Business question
Stored procedures with dynamic SQL	sp_top_areas and sp_compare_cities_by with whitelisted columns	Self-service ranking without SQL injection

Dynamic filtering is handled in two ways. Inside MySQL, a prepared statement accepts a city, a room type and a price band, and NULL means all. For example, CALL sp_top_areas('LON', 'revenue', 10) returns the ten London boroughs with the highest estimated revenue, and the same call with 'occupancy' re-ranks them. The column used for sorting is chosen from a fixed list inside the procedure, so user input never becomes part of the SQL text. In Power BI, slicers apply the same filters visually.

4.2 KPI Framework

A KPI is only useful if everyone calculates it the same way. Each KPI is therefore defined once in SQL and once as a DAX measure with the same logic, and two of them carry an explicit target so that the dashboard can show performance rather than just numbers.

Table 7: KPI definitions used in SQL and Power BI

KPI	Definition	Target or use
Total listings	Count of listings in the snapshot	Market size
Active %	Listings with a valid price, available days and at least one review in the last 12 months, divided by all listings	Target 50%
Median price (USD)	Median nightly price of priced, non-outlier listings	Pricing position
Estimated occupancy %	Booked nights from the review model divided by 365, capped at 70%	Target 15% on the sample
Estimated revenue (USD)	Price multiplied by estimated booked nights, last 12 months	Revenue pool
Long-stay %	Share of listings with a minimum stay of 30 nights or more	Regulation exposure
Top 10% host share	Share of listings held by the largest 10% of hosts	Concentration risk
Last refreshed	Latest timestamp in kpi_snapshot	Proof of live data

Note: the 50% and 15% targets are management benchmarks set for this project, not industry standards.

5. Power BI Dashboard and Storytelling

5.1 Connection and Data Model

Power BI connects to MySQL through the native MySQL connector in Import mode, since DirectQuery is not available for that source. Instead of raw tables, the report reads the vw_pbi_* views, which already hold clean column names and sort orders. In the model the tables were renamed to business names (City, Neighbourhood, RoomType, Listing, Reviews, KPISnapshot and DataQuality), auto-detected relationships that created ambiguous paths were removed, and the remaining one-to-many links were activated. Thirteen DAX measures sit on top of this model.

5.2 Dashboard Pages

The report follows a simple storyline: start with the headline comparison, move to where the business sits on the map, open a single neighbourhood, and finish with whether the KPIs are on target and up to date (Few, 2013; Knaflic, 2015).

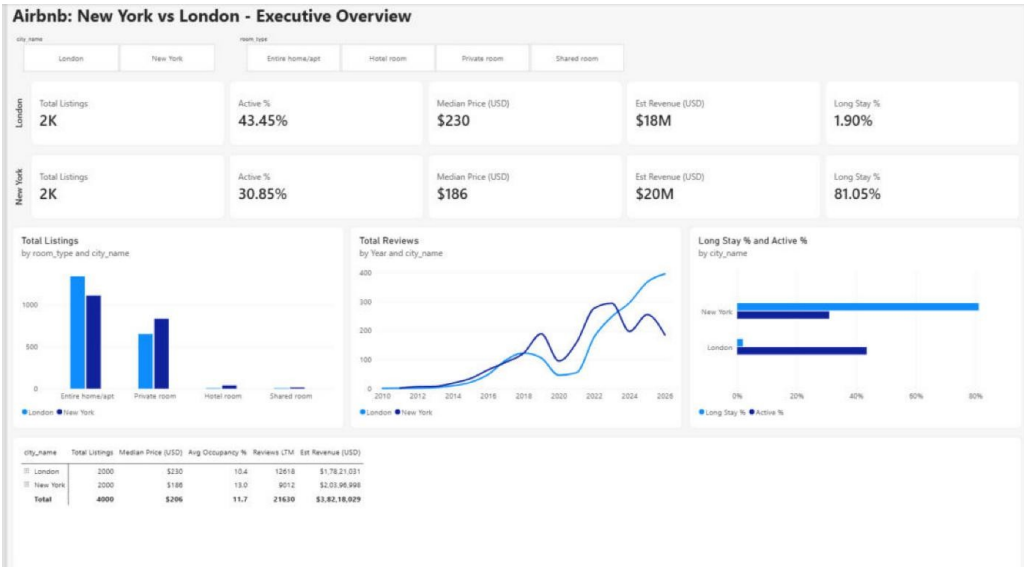


Figure 3: Overview page with city and room-type slicers, KPI cards by city, trend charts and a drill-down matrix

The Overview page answers the first question a manager asks: how do the two cities compare today? The KPI cards are split by city, so the gap in long-stay share (81% against 2%) is visible at a glance. The review trend shows the two markets moving together until 2023 and then separating.

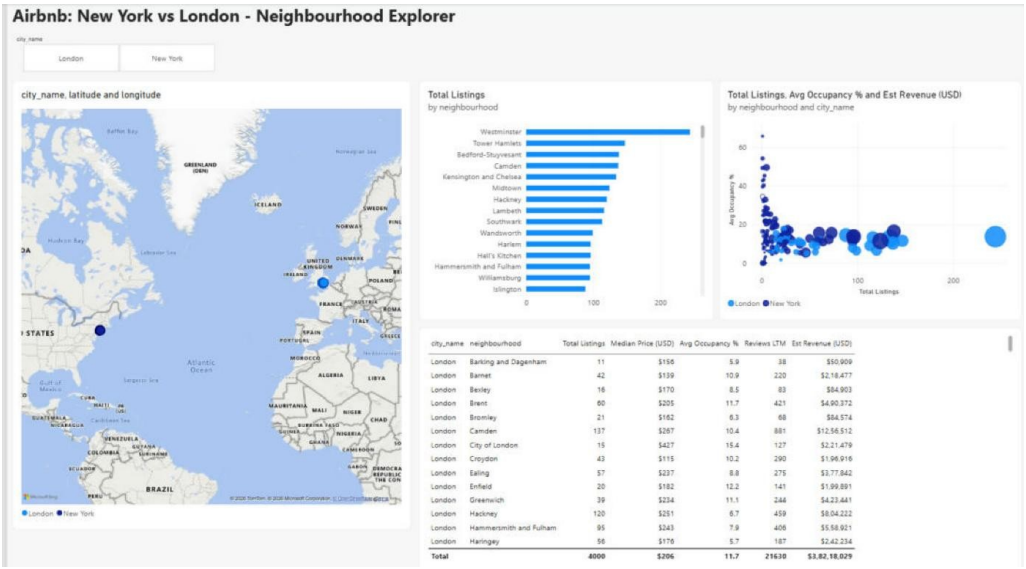


Figure 4: Neighbourhoods page with map, ranking, opportunity scatter and detail table

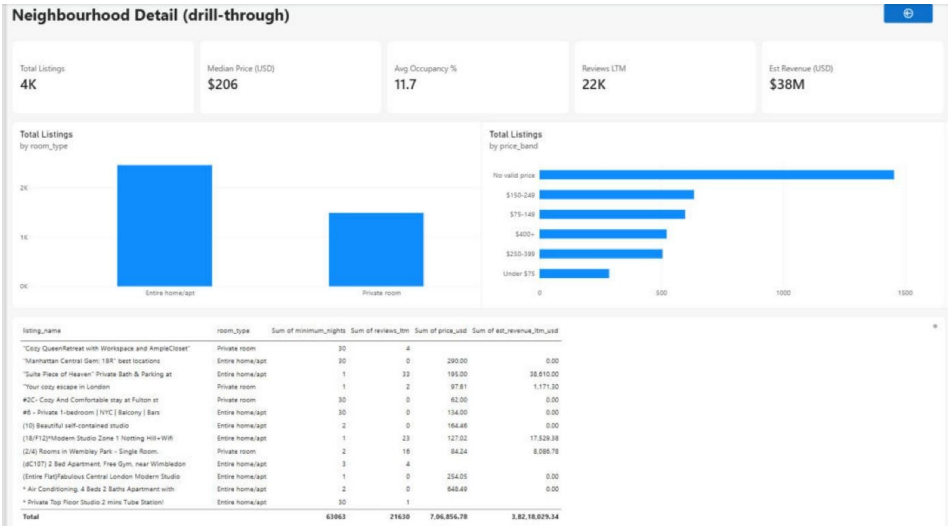


Figure 5: Neighbourhood Detail drill-through page (shown unfiltered)

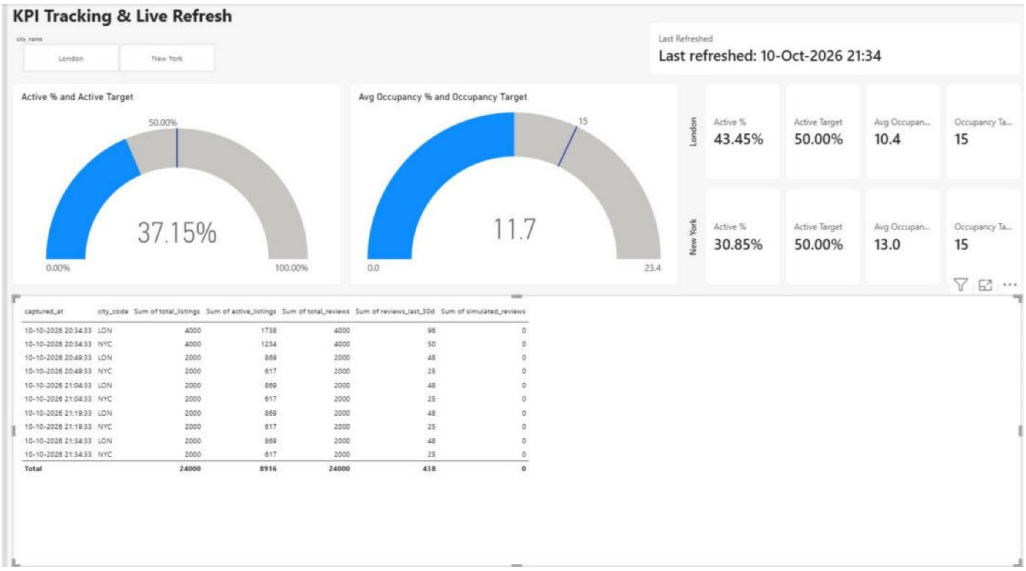


Figure 6: KPI Tracking page with targets, a last-refreshed card and the 15-minute snapshot log

The KPI Tracking page is where the real-time element becomes visible. The gauges compare Active % and occupancy with their targets, the city cards show that both markets fall short of the 50% activity target, and the snapshot table lists every 15-minute capture written by the MySQL event.

5.3 Filters, Drill-Down, Drill-Up and Drill-Through

Interactivity was designed around three different ways of moving through the data, and it helps to keep them apart. Filters, in the form of slicers for city and room type, narrow every visual on the page at once. Drill-down and drill-up work inside a single visual: in the summary matrix on the Overview page, a click on the plus sign beside London opens its room types, and the minus sign or the drill-up arrow closes them again, so the manager moves from city level to segment level without leaving the table. Drill-through moves to another page. On the Neighbourhoods page, right-clicking any neighbourhood in the bar

chart, the scatter or the table and choosing Drill through opens the Neighbourhood Detail page, already filtered to that area, with its own KPIs, room mix, price bands and listing list. The back button returns to the starting page (Microsoft, 2025).

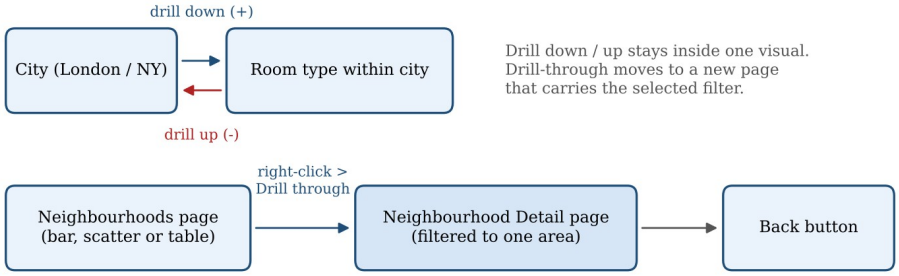


Figure 7: How drill-down, drill-up and drill-through are used in the report

6. Findings and Analysis

6.1 Market Scale and Activity

Table 8: City scorecard, full dataset, September 2026

Indicator	New York	London	Comment
Listings	29,918	91,954	London is 3.1 times larger
Hosts	16,148	51,348	Similar listings per host (1.85 and 1.79)
Active listings	9,265 (31.0%)	39,824 (43.3%)	Less than half the supply is truly trading
Entire homes	54.8%	66.4%	New York relies more on private rooms
Median nightly price (USD)	182	234	London prices are higher overall
Minimum stay of 30+ nights	80.7%	2.1%	The clearest effect of regulation
Reviews in last 12 months per listing	4.9	6.4	Demand per listing is 31% higher in London
Estimated revenue, last 12 months	$316.6 million	$795.7 million	London is 2.5 times larger
Revenue per active listing	about $34,200	about $20,000	An active New York listing earns 1.7 times more

Source: author's SQL queries Q1, Q7 and Q15 on the full dataset.

The first finding is that headline listing counts overstate both markets. Only 31% of New York listings and 43% of London listings are active, meaning they show a price, open days and at least one review in the past year. In New York a third of all listings (33.9%) are dormant: they show a price and open dates but have not had a review in a year. These listings make the platform look bigger than it is and can frustrate guests who try to book them.

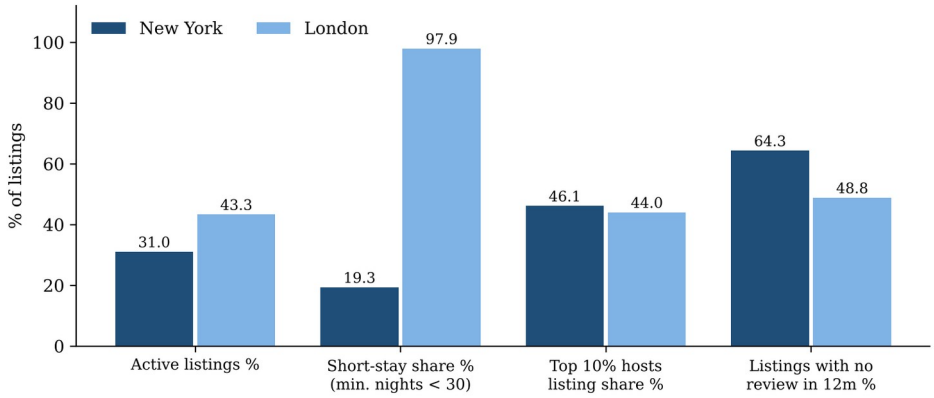


Figure 8: Activity, regulation and concentration indicators by city (% of listings)

6.2 Regulation and Demand

Figure 9 indexes rolling twelve-month review counts to August 2023, just before New York started enforcing Local Law 18. The two lines were almost identical until then. Within a year New York had fallen to 66 while London had risen to 127. New York has recovered slowly to 95, but London stands at 226, more than double its 2023 level. The supply data explains why. Among New York listings that still accept short stays, 2,188 are registered with the city and 2,826 are exempt, but 757 have no licence; 513 of those are active and generated 17,075 reviews in the last year. That is a compliance risk the platform can measure directly.

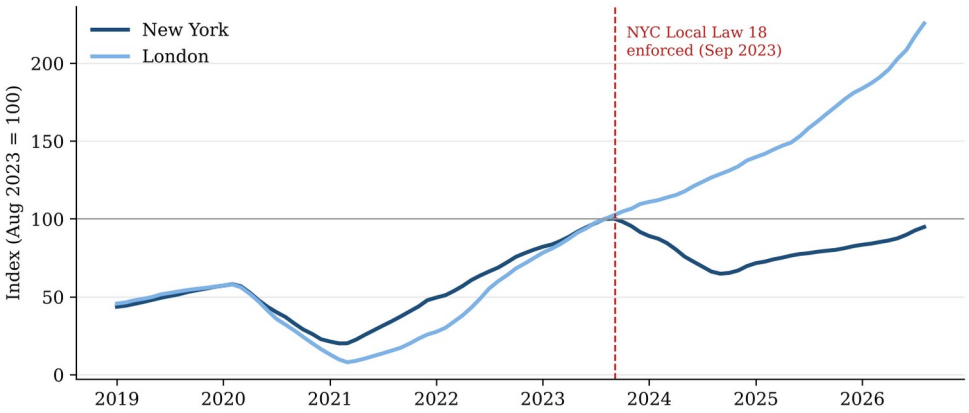


Figure 9: Rolling twelve-month reviews, indexed to August 2023 = 100

6.3 Pricing, Hosts and Neighbourhoods

In both cities the most expensive quarter of listings earns the most, but not because it is booked more often. In New York the top price quartile earns about $70,000 a year per listing against $10,000 in the bottom quartile, while its estimated occupancy is slightly lower. Supply is also concentrated. The largest 10% of hosts hold 46.2% of New York listings and 44.0% of London listings, and they take 55.5% and 66.9% of estimated revenue respectively. The single biggest New York host, a corporate furnished-rental operator, lists 889 homes, all with a 30-night minimum. Revenue is concentrated by place as well: 11 of London's 33

boroughs produce 80% of its revenue, and 35 of New York's 206 active neighbourhoods do the same. Westminster alone earns about $180 million, matching the whole of Manhattan.

The neighbourhood opportunity matrix (query Q13) compares each area's supply with its occupancy. In Brooklyn, areas such as Park Slope, Fort Greene and Clinton Hill run at 43% to 46% estimated occupancy with only 200 to 270 listings each, which suggests demand is not being met. In London, Haringey, Newham and Greenwich play the same role, while Wandsworth has many listings but below-average occupancy.

7. Comparison with Competitors

Airbnb does not operate in a vacuum. Booking Holdings and Expedia Group both sell holiday homes alongside hotels, and in a city such as New York, where most short stays are restricted, guests can simply book a hotel instead. Table 9 compares the three companies on their most recent full-year results.

Table 9: Airbnb and its main online competitors, full year 2025

Measure	Airbnb	Booking Holdings	Expedia Group (incl. Vrbo)
Revenue	$12.2 billion (+10%)	$26.9 billion (+13%)	$14.7 billion (+8%)
Gross bookings	$91.3 billion (+12%)	$186.1 billion (+12%)	$119.6 billion (+8%)
Nights booked	533 million nights and seats	1,235 million room nights	415 million room nights
Core model	Homes and experiences, peer hosts	Hotels first, growing alternative stays	Hotels, flights and Vrbo whole homes
Exposure to city rules	High: most supply is private homes	Lower: hotel inventory dominates	Medium: Vrbo leans to holiday areas, not city centres

Sources: Short Term Rentalz (2026) for Airbnb; Booking Holdings (2026); Expedia Group (2026).

Two points matter for the regional team. Booking Holdings is more than twice Airbnb's size by revenue and is pushing into alternative accommodation, so any London host who becomes unhappy with Airbnb has an obvious alternative. And because New York rules have shifted short stays back towards hotels, Airbnb's advantage there is weakest exactly where Booking and Expedia are strongest. Airbnb's distinctive asset is its long-stay and experiences offer, which neither rival matches at the same scale.

8. Managerial Recommendations

Table 10: Prioritised recommendations

Priority	Recommendation	Evidence	KPI to watch
1	New York: build a mid-term stay product for 30 nights and more, aimed at corporate and relocation guests, with flexible pricing and monthly billing.	80.7% of supply is already 30+ nights; short-stay demand fell by a third	Mid-term bookings; revenue per active listing

Priority	Recommendation	Evidence	KPI to watch
1	New York: contact the 757 unlicensed short-stay hosts and either help them register or move them to 30-night minimums.	513 are active with 17,075 reviews	Unlicensed short-stay listings
1	London: protect the core boroughs (Westminster, Camden, Kensington and Chelsea) through host quality support and help with the 90-night rule.	11 boroughs make 80% of revenue	Active %; reviews per listing
2	Reactivate or remove dormant listings with pricing and calendar nudges, and hide those idle for more than a year from search.	33.9% (NYC) and 21.5% (LON) are open but unreviewed for 12 months	Active % against the 50% target
2	Recruit hosts in under-supplied, high-occupancy areas such as Park Slope, Fort Greene, Haringey and Newham.	43% to 46% occupancy with few listings	New listings by area
2	Create an account-management programme for the top 10% of hosts, with service standards and early warning of churn.	They hold about 45% of listings	Top-host share; host churn
3	Use the long-stay and experiences offer to compete with Booking and Expedia rather than on price alone.	Competitors dominate hotel inventory	Share of nights over 7 days
3	Move the pipeline to the cloud (S3, Spark, Airflow, Power BI Service) with hourly refresh and alerts on KPI thresholds.	The MySQL design already mirrors those layers	Data latency; refresh failures

The recommendations follow directly from the numbers. Because a New York listing that is active earns about $34,000 a year, bringing even 1,000 dormant listings back into use, if they performed like today's active ones, would be worth roughly $34 million in host revenue. Priority reflects speed of action and revenue protected.

9. Limitations and Conclusion

9.1 Limitations

Occupancy and revenue are estimates based on reviews, not on real bookings, so they should be read as relative rather than exact values. About 35% of listings had no price in the snapshot and were left out of price measures. The live demonstration runs on a 2,000-listing sample per city, and the refresh is micro-batch rather than streaming.

9.2 Conclusion

MySQL and Power BI were enough to reproduce the main layers of a big data analytics platform: raw data lands first, transformation and quality checks run in the database, KPIs are captured on a schedule and managers explore them interactively. The analysis itself tells a clear story. London is a large, active, short-stay market whose risk lies in concentration and in competition from Booking. New York has become a mid-term rental market since Local Law 18, smaller in volume but higher in value per active listing. A single strategy will not suit both, and the dashboard gives the regional team the evidence to treat them differently.

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